#2.5

Let P = # of plain, G = # of glazed, C = # of chocolate

$P+G+C=12$

donut, variety

$$\binom{12+3-1}{3-1}=\binom{14}{2}=\frac{14!}{2!\,12!}=\frac{14\cdot 13}{2}=91$$

91

2.17

a) $P(\text{item is bad}) = \frac{\#\text{ bad items}}{\#\text{ items}} = \frac{(2+1)}{(2+1+6+9)} = \frac{3}{18} = \frac{1}{6}$

(bad, good)

$P(\text{item is bad}) = \frac{1}{6}$

b) $P(\text{item is bad} \mid \text{from Assembly 1}) = \frac{\#\text{ bad items from line 1}}{\#\text{ items from line 1}}$

$\frac{(2)}{(2+6)} = \frac{2}{8} = \frac{1}{4}$

$P(\text{item is bad} \mid \text{from line 1}) = \frac{1}{4}$

c) $P(\text{item from line 1} \mid \text{item is bad}) = \frac{\#\text{ bad items from line 1}}{\#\text{ bad items}}$

$= \frac{2}{(1+2)}$

$P(\text{item from line 1} \mid \text{item is bad}) = \frac{2}{3}$

2.19) Show that if $P(A|B) = P(A)$, then $P(B|A) = P(B)$ (This proves Lemma 2.2.1)

- If $P(A|B) = P(A)$, A and B are independent events.

By the multiplication rule, $P(A \cap B) = P(A|B)\,P(B)$, and by substitution, when A and B are independent, $P(A \cap B) = P(A)\,P(B)$.

Considering $P(B|A) = \frac{P(A \cap B)}{P(A)}$, and again using substitution,

$P(B|A) = \frac{P(A)\,P(B)}{P(A)} = P(B)$.

2.20 Use definition of independent events to show that:

(a) if A and B are independent, then A and B^c are also independent

▷ Since $P(A \cap B) = P(A)P(B)$,

▷ Then $P(A \cap B^c) = P(A) - P(A \cap B) = P(A) - P(A)P(B) = P(A)(1 - P(B)) = P(A)P(B^c)$

Q5) Exercise 2.25

a) $P(M \text{ or } Th) = P(M) + P(Th)$

$= 0.20 + 0.27 = \boxed{0.47}$

b) D = defective

$P(D) = (0.2)(0.02) + (0.25)(0.015) + (0.28)(0.01) + (0.27)(0.03)$

$= 0.004 + 0.00375 + 0.0028 + 0.0081 = 0.01865$

$P(M \text{ or } Th \mid D) = \frac{(0.2)(0.02) + (0.27)(0.03)}{0.01865} = \frac{0.0121}{0.01865} = \boxed{0.6488}$

c) π = function properly

$P(\pi) = 1 - P(D) = 1 - 0.01865 = 0.98135$

For Monday or Thursday and function properly

$(0.20)(0.98) + (0.27)(0.97) = 0.196 + 0.2619 = 0.4579$

So, $P(M \text{ or } Th \mid \pi) = \frac{0.4579}{0.98135} \approx \boxed{0.466}$

2.31

(a) ~~P(reject | 10 defective)~~

$P(\text{all 4 is good}) = \frac{90}{100} \cdot \frac{89}{99} \cdot \frac{88}{98} \cdot \frac{87}{97}$

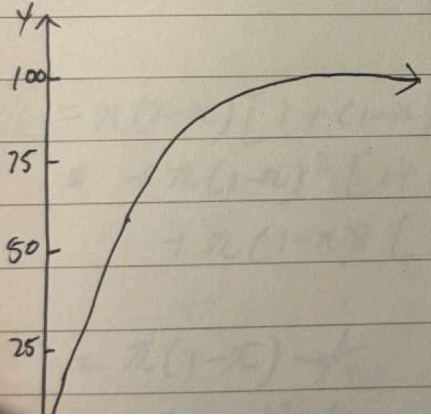
$P(\text{reject}) = 1 - P(\text{all 4 is good}) = 0.3484$

Even if the defection rate is 10%, the rejection rate could still be 34.84%.

(b) $y = 100\left(1 - \frac{(100-x)(99-x)(98-x)(97-x)}{100 \cdot 99 \cdot 98 \cdot 97}\right)$

y = probability of rejection if draws 4 time out of 100 production

x = number of defection in 100 production



2.37 8 blue socks 4 brown 5 blue

how many 3 combination

$\binom{17}{3} = 680$

(a) a matching pair means not a color of each

$p(\text{a color of each}) = \frac{\binom{8}{1}\cdot\binom{5}{1}\cdot\binom{4}{1}}{\binom{17}{3}} = \frac{8\cdot 5\cdot 4}{680} = \frac{4}{17} = 0.2353$

$p(\text{match}) = 1 - \frac{4}{17} = \frac{13}{17} = 0.7647$

(b) p(a pair of black socked)

exactly 2 black and 1 other $\binom{8}{2}\cdot\binom{9}{1} = 252$

all 3 are black $= \binom{8}{3}$

$p(\text{two pair of black}) = \frac{\binom{8}{2}\cdot\binom{9}{1} + \binom{8}{3}}{\binom{17}{3}} = \frac{252+56}{680}$

$= \frac{77}{170} \approx 0.4529$

2.43

x	1	2	3	4
P(X=x)	4/10	3/10	2/10	1/10

$1 \to \frac{2}{5}$

$2 \to \frac{3}{5}\cdot\frac{2}{4} = \frac{6}{20} = \frac{3}{10}$

$3 \to \frac{3}{5}\cdot\frac{2}{4}\cdot\frac{2}{3} = \frac{12}{60} = \frac{2}{10}$

$4 \to \frac{3}{5}\cdot\frac{2}{4}\cdot\frac{1}{3}\cdot\frac{2}{2} = \frac{12}{120} = \frac{1}{10}$